FINALS SCOPE (CH8–CH11)

What Gets Tested Most

Theme	Must-do skill
Hypothesis tests (every final)	choose case → TS → p-value → decision
Two-sample means	pooled vs Welch vs paired
CI ↔ test link	null outside CI ⇒ reject H_0
GOF χ^2	E_i, then χ^2, then $df = k-1-m$
Regression (every final)	test β_1, ANOVA, CI vs PI
Residual/QQ (common)	curve, funnel, outlier, tail bends

See Test Selector table (pg 2) for full Case 1–11 decision guide.

PRE-CH8 CORE (CONDENSED)

Probability + Bayes

$0 \le P(A) \le 1$, $P(A^c) = 1 - P(A)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(B|A) = \frac{P(A \cap B)}{P(A)}$, indep $\Leftrightarrow P(A \cap B) = P(A)P(B)$

Total prob: $P(A) = \sum_i P(A|B_i)P(B_i)$

Bayes: $P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$ *Ex:* $P(A|B) = 0.8$, $P(B) = 0.3$, $P(A) = 0.5 \Rightarrow P(B|A) = 0.48$

PMF/PDF/CDF

Discrete: PMF $f(x)$, sums. Continuous: PDF $f(x)$, integrals.

$F(x) = P(X \le x)$, $P(a < X \le b) = F(b) - F(a)$, $f(x) = F'(x)$ (continuous)

Valid PDF: $f(x) \ge 0$, $\int f(x)\,dx = 1$

Discrete vs Continuous Quick

	Discrete	Continuous
Prob fn	PMF $f(x)$	PDF $f(x)$
$P(X = a)$	can be > 0	$= 0$
$P(X > x)$	$\sum_{t>x} f(t)$	$\int_x^\infty f(t)\,dt$
Mean	$\sum x f(x)$	$\int x f(x)\,dx$

CDF checks: non-decreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$.

Joint / Conditional / Independence

Discrete marginals: $g(x) = \sum_y f(x,y)$, $h(y) = \sum_x f(x,y)$

Continuous marginals: $g(x) = \int f(x,y)\,dy$, $h(y) = \int f(x,y)\,dx$

$f(x|y) = \frac{f(x,y)}{f(y)}$

Indep $\Leftrightarrow f(x,y) = g(x)h(y)$ for all (x,y)

$P(a \le X \le b,\ c \le Y \le d) = \int_c^d \int_a^b f(x,y)\,dx\,dy$

E, Var, Cov, Corr

$E[X] = \sum x f(x)$ or $\int x f(x)\,dx$

$E[aX + b] = aE[X] + b$, $E[X + Y] = E[X] + E[Y]$

$\mathrm{Var}(X) = E[X^2] - E[X]^2$, $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$

$\mathrm{Var}(aX + bY) = a^2V_X + b^2V_Y + 2ab\,\mathrm{Cov}(X,Y)$

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$

$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X,Y)$

Indep $\Rightarrow$ Cov $= 0$ (but Cov $= 0 \not\Rightarrow$ indep)

$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$

Ch5: Discrete Dist Quick

Situation	Distribution
Fixed n, # successes	Binomial
Fixed n, k categories	Multinomial
No replacement	Hypergeometric
# trials until k-th success	Neg. Binomial
# trials until first success	Geometric
# events in interval	Poisson

Bin(n,p): $P = \binom{n}{x}p^xq^{n-x}$, $\mu = np$, $\sigma^2 = npq$

Hypergeo(N,K,n): $P = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$, $\mu = \frac{nK}{N}$

NB(k,p): $P = \binom{x-1}{k-1}p^kq^{x-k}$, $\mu = \frac{k}{p}$, $\sigma^2 = \frac{kq}{p^2}$

Geo(p): $P = pq^{x-1}$, $\mu = \frac{1}{p}$, $\sigma^2 = \frac{q}{p^2}$, memoryless

Poi(λ): $P = e^{-\lambda}\lambda^x/x!$, $\mu = \sigma^2 = \lambda$

$q = 1 - p$ everywhere. Approx: Bin$(n,p) \approx$Poi(np) if large n, small p

With replacement ⇒ Binomial model; without replacement ⇒ Hypergeometric model.

Ch5 Extras (Useful in Word Problems)

Binomial wording: "exactly x" uses $\binom{n}{x}p^xq^{n-x}$; fixed sequence uses only p^xq^{n-x}.

Geo tail: $P(X > k) = q^k$, memoryless $P(X > s + t|X > s) = P(X > t)$.

Acceptance sampling (ship if $X \le c$ defectives): $P(\text{ship}) = \sum_{x=0}^{c} \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$.

Poisson additivity: if $X_i \sim$Poi(λ_i) indep, then $\sum X_i \sim$Poi$(\sum \lambda_i)$.

Ch6: Continuous Dist Quick

$X \sim N(\mu, \sigma^2)$: $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

Normal approx Bin(n,p): $Y \sim N(np, np(1-p))$, use continuity correction

Normal approx Poi(λ): $Y \sim N(\lambda, \lambda)$ (use continuity correction) *Ex: Binomial CC:* $P(X \le 10) \approx P(Y \le 10.5)$

Uniform(A,B): $f(x) = \frac{1}{B-A}$, $\mu = \frac{A+B}{2}$, $\sigma^2 = \frac{(B-A)^2}{12}$

Exp(β): $f = \frac{1}{\beta}e^{-x/\beta}$, $\mu = \beta$, $\sigma^2 = \beta^2$, $P(X > t) = e^{-t/\beta}$, memoryless

Gamma(α,β): $f = \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha\Gamma(\alpha)}$, $\mu = \alpha\beta$, $\sigma^2 = \alpha\beta^2$

$\chi^2(\nu) =$Gamma$(\nu/2, 2)$: $\mu = \nu$, $\sigma^2 = 2\nu$

Weibull(α,β): $F(x) = 1 - e^{-\alpha x^\beta}$, hazard $z(t) = \alpha\beta t^{\beta-1}$ **Weibull hazard: $\beta < 1$ decreasing, $\beta = 1$ constant, $\beta > 1$ increasing.**

Poisson process: count in $t \sim$Poi(λt), interarrival$\sim$Exp$(1/\lambda)$, time to kth$\sim$Gamma$(k, 1/\lambda)$

Distribution Summary (Pre-Ch8)

Dist	Mean	Var
Bin(n,p)	np	npq
Poi(λ)	λ	λ
Geo(p)	$1/p$	q/p^2
Exp(β)	β	β^2
Unif(A,B)	$(A+B)/2$	$(B-A)^2/12$

Supports: Bin $0..n$, Poi $0, 1, \ldots$, Geo $1, 2, \ldots$, Exp $x > 0$, Unif $A \le x \le B$.

Normal / Z Table Cues

$P(Z > z) = 1 - \Phi(z)$, $P(Z < -z) = P(Z > z)$, $P(|Z| < z) = 2\Phi(z) - 1$.

$P(a < Z < b) = \Phi(b) - \Phi(a)$, $P(|Z| > z) = 2[1 - \Phi(z)]$.

For $X \sim N(\mu, \sigma^2)$: standardize with $Z = \frac{X-\mu}{\sigma}$ before table lookup.

Conditional Expectation

$E(Y|X = x) = \int y\,f(y|x)\,dy = \int y\,\frac{f(x,y)}{g(x)}\,dy$

$E(Y) = E[E(Y|X)] = \int E(Y|x)g(x)\,dx$ (iterated E)

$\mathrm{Var}(Y) = E[\mathrm{Var}(Y|X)] + \mathrm{Var}(E[Y|X])$ (law of total var)

Reliability + Hazard

$R(t) = P(T > t) = 1 - F(t)$, $z(t) = \frac{f(t)}{R(t)}$.

$f(t) = z(t)R(t)$, $R(t) = \exp\left(-\int_0^t z(u)\,du\right)$.

Exp(β) has constant hazard $z(t) = 1/\beta$.

Ch7: Transforms

1-to-1: $g(y) = f(w(y))\left|\frac{dw}{dy}\right|$, w = inverse of u

CDF method: $F_Y(y) = P(u(X) \le y)$, then $f_Y = F_Y'$

2D Jacobian: $f_{U,V}(u,v) = f_{X,Y}(x(u,v), y(u,v)) \cdot |J|$

$$J = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix}$$

Linear Comb of Normals

If $X_i \sim N(\mu_i, \sigma_i^2)$ indep, then $\sum a_iX_i \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$.

IID normal: $\sum X_i \sim N(n\mu, n\sigma^2)$, $\bar{X} \sim N(\mu, \sigma^2/n)$.

Transform Shortcuts

If $Y = aX + b$: $E[Y] = aE[X] + b$, $\mathrm{Var}(Y) = a^2\mathrm{Var}(X)$.

If $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$.

If $X \sim$Exp(β) and $Y = cX$ $(c > 0)$, then $Y \sim$Exp$(c\beta)$.

MGF

$M_X(t) = E(e^{tX})$, $E[X] = M'(0)$, $E[X^2] = M''(0)$

$\mathrm{Var} = M''(0) - [M'(0)]^2$

If indep: $M_{X_1+\cdots+X_n}(t) = \prod_i M_{X_i}(t)$

MGF uniquely determines distribution.

Common MGFs

Dist	MGF $M(t)$
Bin(n,p)	$(pe^t + q)^n$
Poi(λ)	$e^{\lambda(e^t-1)}$
Geo(p)	$\frac{pe^t}{1-qe^t}$
$N(\mu,\sigma^2)$	$e^{\mu t+\sigma^2t^2/2}$
Exp(β)	$(1-\beta t)^{-1}$
Gamma(α,β)	$(1-\beta t)^{-\alpha}$
$\chi^2(\nu)$	$(1-2t)^{-\nu/2}$

Sum of indep normals is normal. Sum of indep Poi is Poi. Sum of indep χ^2 is χ^2. Recognize MGF → identify distribution. *Ex:* $M(t) = e^{3(e^t-1)} \Rightarrow Poi(3)$. $M(t) = (0.4e^t + 0.6)^{10} \Rightarrow Bin(10, 0.4)$.

Useful Sums + Approx Rules

$\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

$\sum_{i=0}^\infty x^i = \frac{1}{1-x}$ $(|x| < 1)$, $\sum_{i=1}^\infty ix^{i-1} = \frac{1}{(1-x)^2}$

Bin→Normal: $np \ge 5$, $nq \ge 5$ (use CC).

Poi→Normal: $\lambda \gtrsim 10$ (use CC).

Hypergeo→Bin: if sampling fraction $n/N < 0.05$, use Bin$(n, p = K/N)$.

CH8: SAMPLING DISTRIBUTIONS

Sampling Basics

$\bar{X} = \frac{1}{n}\sum X_i$, $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$

σ^2 = **population** variance (parameter, fixed, usually unknown)

S^2 = **sample** variance (statistic, random, estimate of σ^2)

σ given ⇒ use Z. s given ⇒ use t.

Normal Sampling & CLT

$X_i \sim N(\mu, \sigma^2)$: $\bar{X} \sim N(\mu, \sigma^2/n)$ **exactly**

CLT: any dist, $n \ge 30$: $\bar{X} \overset{\text{approx}}{\sim} N(\mu, \sigma^2/n)$

$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ (σ known)

$\sum X_i \sim N(n\mu, n\sigma^2)$

Two-sample: $\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$

Chebyshev & Markov

Markov ($X \ge 0$, only need $E[X]$): $P(X \ge a) \le \frac{E[X]}{a}$

Chebyshev (need μ and σ^2, **any** dist): $P(\mu - k\sigma < X < \mu + k\sigma) \ge 1 - \frac{1}{k^2}$

Equiv: $P(|X - \mu| \ge \varepsilon) \le \frac{\sigma^2}{\varepsilon^2}$ (set $\varepsilon = k\sigma$)

χ^2, Independence of $\bar{X}$ & S^2

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$ (normal pop)

$\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$

Additivity: $\chi^2(\nu_1) + \chi^2(\nu_2) \sim \chi^2(\nu_1 + \nu_2)$

$\bar{X}$ and S^2 are independent (normal pop only!)

t-Distribution

$T = \frac{Z}{\sqrt{V/\nu}} \sim t(\nu)$, $Z \sim N(0,1)$, $V \sim \chi^2(\nu)$ indep

One-sample: $T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$

Reading table: $P(T > t_{\alpha,\nu}) = \alpha$, row=df, col= α

F-Distribution

$F = \frac{U/\nu_1}{V/\nu_2} \sim F(\nu_1, \nu_2)$

Compare variances: $F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$

Under $H_0: \sigma_1^2 = \sigma_2^2$: $F = \frac{S_1^2}{S_2^2}$

$F_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{F_\alpha(\nu_2, \nu_1)}$ (reciprocal trick)

CH9: ESTIMATION

Point Estimation

Unbiased: $E(\hat{\Theta}) = \theta$. $E(\bar{X}) = \mu$, $E(S^2) = \sigma^2$.

SE: $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$. **SE ≠ Std dev!**

How to Read Tables

Z table: gives $P(Z \le z)$. Look up z, read left-tail area.

t table: rows=df, cols= α. Entry $t_{\alpha,\nu}$ is the value with area α to its *right*: $P(T_\nu > t_{\alpha,\nu}) = \alpha$.

χ^2 table: same idea. $\chi^2_{\alpha,\nu}$: area α to the right.

F table: F_{α,ν_1,ν_2}: area α to the right. Need ν_1 (num) and ν_2 (denom) df.

CH9: CONFIDENCE INTERVALS

$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ (known), $E = t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$ (unk). CI: $\bar{X} \pm E$

Width ↑ as confidence↑, σ ↑, n ↓

σ known ⇒ Z. σ unknown ⇒ t.

Common z Values

	90%	95%	99%
2-sided $z_{\alpha/2}$	1.645	1.96	2.576
1-sided z_α	1.282	1.645	2.326

For confidence C, set $\alpha = 1 - C$. Two-sided uses $z_{\alpha/2}$; one-sided uses z_α.

One-Sided CI

Upper: $\mu \le \bar{X} + z_\alpha\frac{\sigma}{\sqrt{n}}$

Lower: $\mu \ge \bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}$

Use z_α not $z_{\alpha/2}$! All α in one tail.

Sample Size for Error e

$n \ge \left(\frac{z_{\alpha/2}\,\sigma}{e}\right)^2$ (round UP)

CI for Proportion p

$\hat{p} = \frac{x}{n}$, x = number of successes in sample, n = total sample size, $\hat{q} = 1 - \hat{p}$

$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$, need $n\hat{p} \ge 5$, $n\hat{q} \ge 5$

Use z (not t): large n needed anyway.

Prediction Interval

σ unk: $\bar{X} \pm t_{\alpha/2,n-1}\,S\sqrt{1 + \frac{1}{n}}$

PI > CI always! Why:

CI uncertainty: just $\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$ (uncertainty in estimating the population mean μ)

PI uncertainty: $\mathrm{Var}(X_{\text{new}} - \bar{X}) = \sigma^2 + \frac{\sigma^2}{n}$ = individual variability around μ + mean estimation error. The extra σ^2 comes from the natural spread of a single future observation.

CI Summary

What	Formula
μ, σ known	$\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
μ, σ unk	$\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$
p	$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$
PI (σ unk)	$\bar{X} \pm t_{\alpha/2,n-1}S\sqrt{1 + 1/n}$

CH9: TWO-SAMPLE CIS

Case 3: $\mu_1 - \mu_2$ (σ's known)

$$(\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2}\underbrace{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}_{\text{SE of } (\bar{X}_1 - \bar{X}_2)}$$

$\mathrm{Var}(\bar{X}_1-\bar{X}_2)=\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}$, so SE is the square root of that. This is a Z interval (known σ_1,σ_2), so no t df is needed.

Case 4: $\mu_1-\mu_2$ (unk equal σ: Pooled t)

Pooled variance estimator:

$$S_p^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$$

$$\mathrm{df}=n_1+n_2-2$$

$$(\bar{X}_1-\bar{X}_2)\pm t_{\alpha/2,\,n_1+n_2-2}\,S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$$

SE for pooled t is $S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$.

Case 5: $\mu_1-\mu_2$ (unk unequal σ: Welch t)

$$(\bar{X}_1-\bar{X}_2)\pm t_{\alpha/2,\,\nu}\underbrace{\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}}_{\text{SE of }(\bar{X}_1-\bar{X}_2)}$$

$$\nu^*\approx\frac{\left(\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1}+\frac{(S_2^2/n_2)^2}{n_2-1}}$$

Use $df=\lfloor\nu^*\rfloor$ (round DOWN).

Case 6: Paired Data (μ_D)

$D_i=X_i-Y_i$, $\bar{D}=\frac{1}{n}\sum D_i$, $S_D^2=\frac{1}{n-1}\sum(D_i-\bar{D})^2$

$\mathrm{Var}(D_i)=\mathrm{Var}(X_i-Y_i)=\sigma_X^2+\sigma_Y^2-2\mathrm{Cov}(X_i,Y_i)$

$\bar{D}\pm t_{\alpha/2,n-1}\frac{S_D}{\sqrt{n}}$, $df=n-1$ $S_D=\sqrt{S_D^2}$ is the sample SD of differences; center is $\bar{D}$.

Two Proportions (p_1-p_2)

$$(\hat{p}_1-\hat{p}_2)\pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}$$

CH9: VARIANCE CIS

Single Variance (σ^2)

Let $X_{\chi^2}=\frac{(n-1)S^2}{\sigma^2}$. Then $X_{\chi^2}\sim\chi^2_{n-1}$.

CI: $\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right)$

Ratio of Variances (σ_1^2/σ_2^2)

Let $X_F=\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$. Then $X_F\sim F_{n_1-1,n_2-1}$.

CI: $\left(\frac{S_1^2/S_2^2}{F_{\alpha/2,\nu_1,\nu_2}},\ \frac{S_1^2}{S_2^2}\cdot F_{\alpha/2,\nu_2,\nu_1}\right)$, $\nu_i=n_i-1$

CH9: MLE

$L(\theta)=\prod f(x_i;\theta)$, $\ell=\ln L$. Solve $d\ell/d\theta=0$.

Bin: $\hat{p}=x/n$. Normal: $\hat{\mu}=\bar{X}$, $\hat{\sigma}^2_{\mathrm{MLE}}=\frac{1}{n}\sum(X_i-\bar{X})^2$

MLE σ^2 uses n (biased); S^2 uses $n-1$ (unbiased).

MLE Properties

- **Consistent:** $\hat{\theta}_{MLE}\xrightarrow{p}\theta$
- **Asymptotically normal:** $\hat{\theta}_{MLE}\approx N\big(\theta,\ [nI(\theta)]^{-1}\big)$
- **Asymptotically efficient:** reaches Cramer-Rao lower bound asymptotically
- **Invariance:** MLE of $g(\theta)$ is $g(\hat{\theta})$

TEST SELECTOR (CH9+10)

What?	Cond	Use
1. One μ	σ known	Z
2. One μ	σ unk	T, $df=n-1$
3. $\mu_1-\mu_2$	σ's known	Z
4. $\mu_1-\mu_2$	unk, equal	Pool T, n_1+n_2-2
5. $\mu_1-\mu_2$	unk, uneq	Welch T, $\lfloor\nu^*\rfloor$
6. Paired	same subj	T on D_i, $n-1$
7. σ^2	normal	χ^2, $n-1$
8. σ_1^2/σ_2^2	normal	F, (n_1-1,n_2-1)
9. p	large n	Z
10. GOF	categ	χ^2, $k-1-m$
11. $\beta_1=0$	linear	T or F

Blue=Z (σ known). **Red**=T (σ unk).

1-Sided vs 2-Sided

	1-sided	2-sided
H_a	$>$ or $<$	$\neq$
crit	z_α/t_α	$z_{\alpha/2}/t_{\alpha/2}$
p-val	tail prob	2xtail
rej	TS$>c_\alpha$	$\lvert$TS$\rvert>c_{\alpha/2}$

Ex: If H_a : $\mu>\mu_0$, do not double p-value (right-tail only).

$\alpha=0.05$ Crit Values

$z_{0.05}=1.645$, $z_{0.025}=1.96$, $z_{0.005}=2.576$

p-value $<\alpha\Rightarrow$ reject. Null $\notin$ CI $\Rightarrow$ reject.

CH10: HYPOTHESIS TESTING

Core Setup

H_0 (status quo), H_a (claim), α, test stat, p-value.

Type I: $\alpha=P(\text{rej }H_0\mid H_0\text{ true})$. Type II: $\beta=P(\text{fail rej}\mid H_0\text{ false})$. Power$=1-\beta$.

Test Statistics (Cases 1–8)

Case 1 (σ known): $Z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$

Case 2 (σ unk): $T=\frac{\bar{X}-\mu_0}{S/\sqrt{n}}$, $df=n-1$

Case 3 (σ's known): $Z=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}$

Case 4 (Pooled t, equal σ): $S_p^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$

$T=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{S_p\sqrt{1/n_1+1/n_2}}$, $df=n_1+n_2-2$

Case 5 (Welch t, unequal σ): $T=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{\sqrt{S_1^2/n_1+S_2^2/n_2}}$, $df=\lfloor\nu^*\rfloor$

$\nu^*=\frac{(S_1^2/n_1+S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1-1)+(S_2^2/n_2)^2/(n_2-1)}$

Case 6 (Paired): $D_i=X_i-Y_i$, $T=\frac{\bar{D}-d_0}{S_D/\sqrt{n}}$, $df=n-1$

Case 7: $\chi^2=\frac{(n-1)S^2}{\sigma_0^2}\sim\chi^2_{n-1}$

Case 8: $F=\frac{S_1^2}{S_2^2}\sim F_{n_1-1,n_2-1}$

Proportion Tests

One proportion:

$$Z=\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$$

Use p_0 (not $\hat{p}$) in SE for testing!

Two proportions $H_0: p_1=p_2$: $\hat{p}=\frac{x_1+x_2}{n_1+n_2}$ (pooled), $Z=\frac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1+1/n_2)}}$

Goodness-of-Fit

$$\chi^2=\sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i},\quad df=k-1-m$$

m = number of estimated parameters; typically need each $E_i\geq 5$. Ex: Normal GOF with $k=6$ bins and estimated (μ,σ): $df=6-1-2=3$.

Type II Error β & Power

1. Find crit value: $\bar{x}_c=\mu_0+z_\alpha\frac{\sigma}{\sqrt{n}}$ (right-sided)

2. Under true μ_a: $\beta=P(\bar{X}<\bar{x}_c\mid\mu_a)=\Phi\left(\frac{\bar{x}_c-\mu_a}{\sigma/\sqrt{n}}\right)$

Power $=1-\beta$. Ex: $H_0:\mu=50$, $\sigma=10$, $n=25$, $\alpha=0.05$. $\bar{x}_c=50+1.645(2)=53.29$. If $\mu_a=55$: $\beta=\Phi(\frac{53.29-55}{2})=\Phi(-0.855)=0.196$, power$=0.804$.

Sample Size for Power

$n=\left(\frac{(z_\alpha+z_\beta)\sigma}{\Delta}\right)^2$, $\Delta=|\mu_a-\mu_0|$. Two-sided: use $z_{\alpha/2}$.

CH11: SIMPLE LINEAR REGRESSION

Model + Estimators

$$Y_i=\beta_0+\beta_1x_i+\varepsilon_i,\quad \varepsilon_i\overset{iid}{\sim}N(0,\sigma^2)$$

$S_{xx}=\sum x_i^2-(\sum x_i)^2/n$, $S_{yy}=\sum y_i^2-(\sum y_i)^2/n$

$S_{xy}=\sum x_iy_i-(\sum x_i)(\sum y_i)/n$

$b_1=\frac{S_{xy}}{S_{xx}}$, $b_0=\bar{y}-b_1\bar{x}$, $\hat{y}=b_0+b_1x$

Error + ANOVA Pieces

$e_i=y_i-\hat{y}_i$, SST=SSR+SSE, MS = SS/df

$R^2=\frac{\mathrm{SSR}}{\mathrm{SST}}=1-\frac{\mathrm{SSE}}{\mathrm{SST}}$ (coeff of determination)

$s^2=\mathrm{MSE}=\frac{\mathrm{SSE}}{n-2}$ (unbiased estimator of σ^2)

Source	df	MS
Regression	1	MSR = SSR
Error	$n-2$	MSE = $\frac{\mathrm{SSE}}{n-2}$
Total	$n-1$	

$F=\frac{\mathrm{MSR}}{\mathrm{MSE}}\sim F_{1,n-2}$, reject $H_0:\beta_1=0$ if $F>F_{\alpha,1,n-2}$

$\mathrm{SSR}=\sum(\hat{y}_i-\bar{y})^2=b_1S_{xy}$

$\mathrm{SSE}=\sum e_i^2=S_{yy}-b_1S_{xy}$

$\mathrm{SST}=\sum(y_i-\bar{y})^2=S_{yy}$

Inference on Slope β_1

$$H_0:\beta_1=0,\qquad T=\frac{b_1-0}{s/\sqrt{S_{xx}}}\sim t_{n-2}$$

$$\beta_1\text{ CI: }b_1\pm t_{\alpha/2,n-2}\frac{s}{\sqrt{S_{xx}}}$$

Equivalent ANOVA test: $F=\frac{\mathrm{MSR}}{\mathrm{MSE}}\sim F_{1,n-2}$, $F=t^2$ (simple regr)

CI vs PI at x_0

$\hat{y}_0=b_0+b_1x_0$. Let $h=\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}$.

CI (mean resp): $\hat{y}_0\pm t_{\alpha/2,n-2}\,s\sqrt{h}$

PI (single obs): $\hat{y}_0\pm t_{\alpha/2,n-2}\,s\sqrt{1+h}$

PI > CI (extra "1" = individual noise)

Correlation

$r=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$, $b_1=r\frac{S_y}{S_x}$. Test $\beta_1=0\Leftrightarrow$ test $\rho=0$.

Lack-of-Fit Test

Need replicates (repeated Y at same X). $k=\#$ distinct x values.

$\mathrm{PESS}=\sum_i\sum_j(y_{ij}-\bar{y}_i)^2$ (pure error), df$=n-k$

SSLF = SSE − PESS, df$=k-2$

$F_{\mathrm{LOF}}=\frac{\mathrm{SSLF}/(k-2)}{\mathrm{PESS}/(n-k)}\sim F_{k-2,\,n-k}$

Reject linear model if $F>F_{\alpha,k-2,n-k}$.

Assumptions (LINE)

Linearity, **I**ndep, **N**ormal errors, **E**qual var. Check: residual plot + QQ.

Curve⇒nonlinear; funnel⇒non-constant var; tail bends in QQ⇒non-normal.

R OUTPUT DECODER (WHAT IT IS)

Command to Meaning

`t.test(x, mu=mu0)`: one-sample mean test. $H_0:\mu=\mu_0$.

`t.test(x,y,var.equal=FALSE)`: two independent means (Welch default). $H_0:\mu_1-\mu_2=0$.

`t.test(x,y,var.equal=TRUE)`: pooled two-sample mean test (assume equal variances).

`t.test(x,y,paired=TRUE)`: paired mean-difference test. $H_0:\mu_D=0$ with $D_i=x_i-y_i$.

`prop.test(x,n)`: one proportion test. $H_0:p=p_0$.

`prop.test(c(x1,x2),c(n1,n2))`: two-proportion test. $H_0:p_1=p_2$.

`cor.test(x,y)`: correlation test. $H_0:\rho=0$.

`lm(y ~ x)`: fits regression line $E[Y|X]=\beta_0+\beta_1x$.

Output Fields (What They Mean)

`estimate`: estimated parameter ($\hat{\mu},\hat{p},r,\hat{\beta}_1$).

`statistic`: test statistic (t,z,F,χ^2).

`parameter`: df.

`p.value`: evidence against H_0; reject if p-value $<\alpha$.

`conf.int`: CI for parameter; if null value not in CI, reject at that confidence level.

Regression Row in `summary(lm)`

`Estimate` = slope/intercept estimate.

`Std. Error` = uncertainty of estimate.

`t value` = Estimate / Std. Error.

`Pr(>|t|)` = p-value for coefficient test (usually $H_0:\beta_j=0$).

`Multiple R-squared` = proportion of variance explained.

R Interpretation Cues

Mean<median ⇒ left-skew. Mean>median ⇒ right-skew.

QQ plot: points near line ⇒ normal plausible; S-shape or tail bends ⇒ non-normal.

Common Final Mistakes

Choosing pooled t when equal-variance assumption is not justified.

Doubling p-value in one-sided tests.

Using $z_{\alpha/2}$ for one-sided CI/test instead of z_α (or t_α).

Saying "fail to reject" means H_0 is proven true.